INCH-POUND

MIL-M-28787/262A
23 March 1994
SUPERSEDING
MIL-M-28787/262(EC)
2 June 1982

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INVERTER, BUFFER, DIGITAL, KEY CODE EMN

*This specification is approved for use by all Departments and Agencies of the Department of Defense.

*The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic buffer inverter module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited.
FSC 5963

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.2001 failures per million operating hours for class I or 0.5983 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

* 3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

* 3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be EMN.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation Initial End-of-life	II II	S S	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	MOhm MOhm
* B1-B18	Input clamp diode voltage, negative	III	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
C1-C18	Output voltage, low level, VDD = 5V	IV	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
D1-D18	Output voltage, high level, VDD = 5V	V	X	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	V
E1-E18	Output voltage, low level, VDD = 15V	VI	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
* F1-F18	Output voltage, high level, VDD = 15V	VII	X	13.5	15.15	13.5	15.15	13.5	15.15	13.5	15.15	13.5	15.15	V
G1-G18	Logic level conversion	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H18	Input current, high level	IX	X	0	200	---	---	0	200	---	---	0	200	nA
J1-J18	Input current, low level	X	X	-30	-45	---	---	-30	-45	---	---	-30	-45	μA
K1	Power supply current, low level	XI	X	0	12	0	12	0	175	0	12	0	291	μA
* K2	Power supply current, high level	XI	X	450	887	450	887	450	850	450	887	0	1035	μA
L1-L18	Transition time (tTLH), CL = 50 pF	XII	S	15	200	15	200	20	220	15	160	20	236	nS
L19-L36	Transition time (tTLH), CL = 150 pF	XII	Q	35	400	35	400	40	440	30	305	45	471	nS
M1-M18	Transition time (tTHL), CL = 50 pF	XIII	S	5	80	5	80	10	89	5	61	10	96	nS

-See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T17:11Z
Check the source to verify that this is the current version before use.

* TABLE I Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M19-M36	Transition time (tTHL), CL = 150 pF	XIII	Q	15	140	15	140	15	155	10	110	20	164	nS
N1-N18	Propagation delay time (tPLH), CL = 50 pF	XIV	S	10	160	10	160	15	175	10	122	15	188	nS
N19-N36	Propagation delay time (tPLH), CL = 150 pF	XIV	Q	15	240	15	240	20	285	15	185	20	284	nS
P1-P18	Propagation delay time (tPHL), CL = 50 pF	XV	S	5	60	5	60	5	66	5	45	5	75	nS
P19-P36	Propagation delay time (tPHL), CL = 150 pF	XV	Q	10	140	15	140	15	155	10	107	20	164	nS
Q1-Q18	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
R1-R18	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
T	Power supply filter capacitance	XIX	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
U1-U18	Input capacitance	XX	Q	2	25	2	25	2	25	2	25	2	25	pF

-See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

NOTES:

1/ "X" parameters shall be 100 percent tested during the module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:11Z
Check the source to verify that this is the currentversion before use.

* 3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/262-1 for class I and M28787/262-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.8 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (VDD) of 3 to 18 Vdc ± 5 per cent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:11Z
Check the source to verify that this is the current version before use.

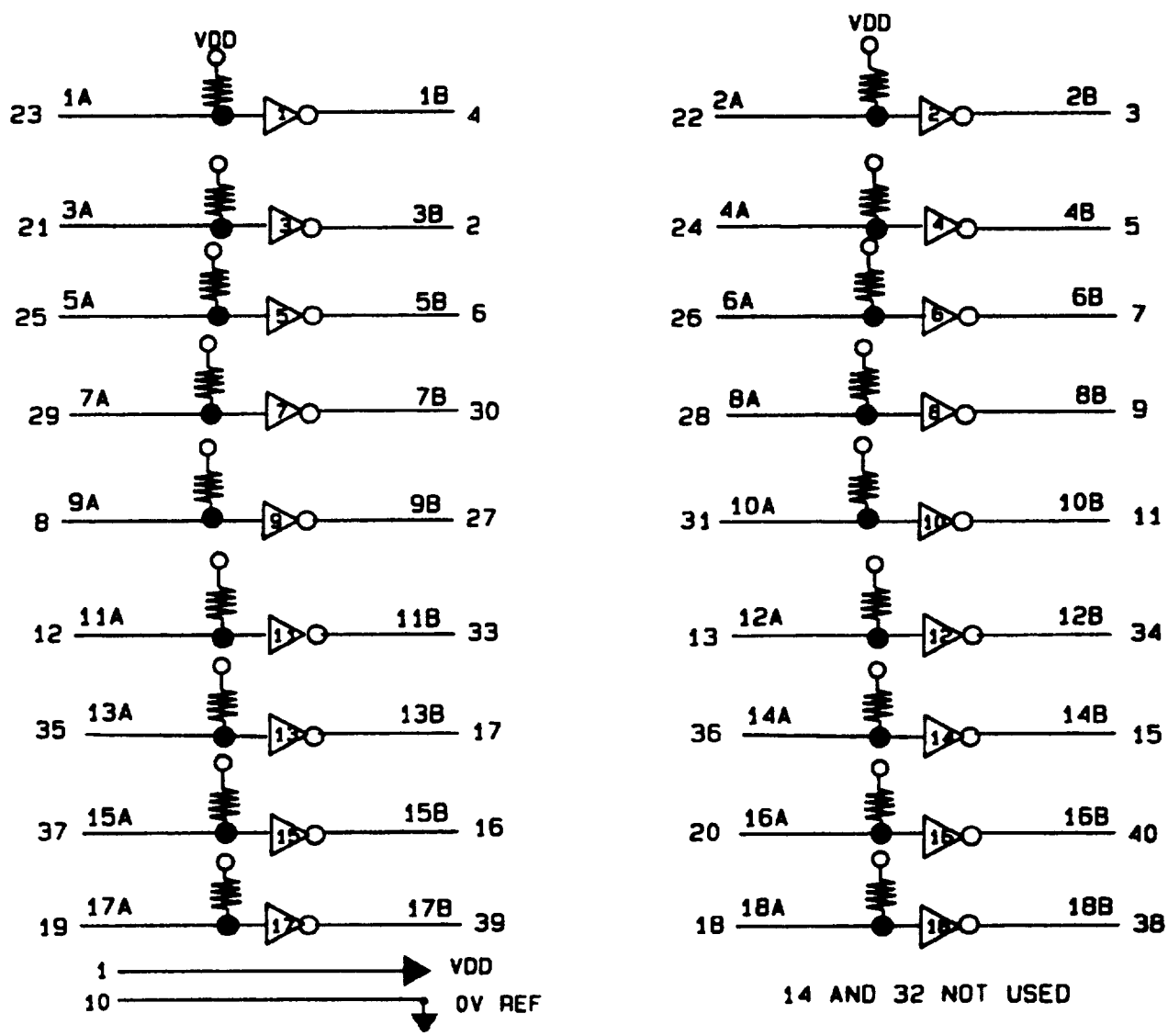


FIGURE 1. EMN module function diagram, buffer inverter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. Test conditions A and B are stress tests and shall be performed only at 25 ± 5°C. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A and B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

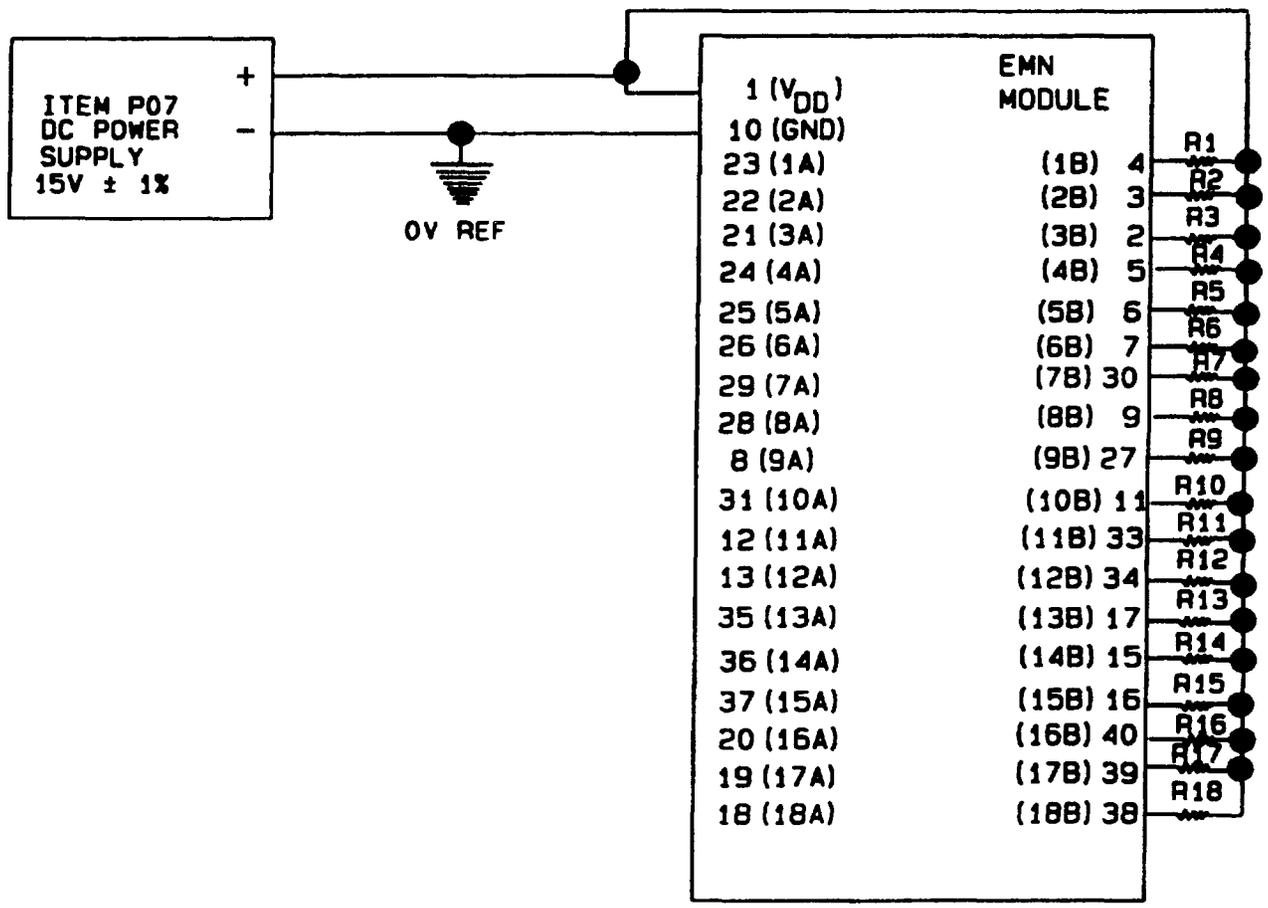
* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through G and K through T shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2. Test conditions H and J shall be performed at 25°, 60° and 85°C only using the initial limits for the applicable operating temperature.

4.5.1.1 Transient thermal test. For the transient thermal test, use the same test conditions as the maximum class temperature test conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition T, shall be performed using the initial limits at 25 ± 5°C ambient.

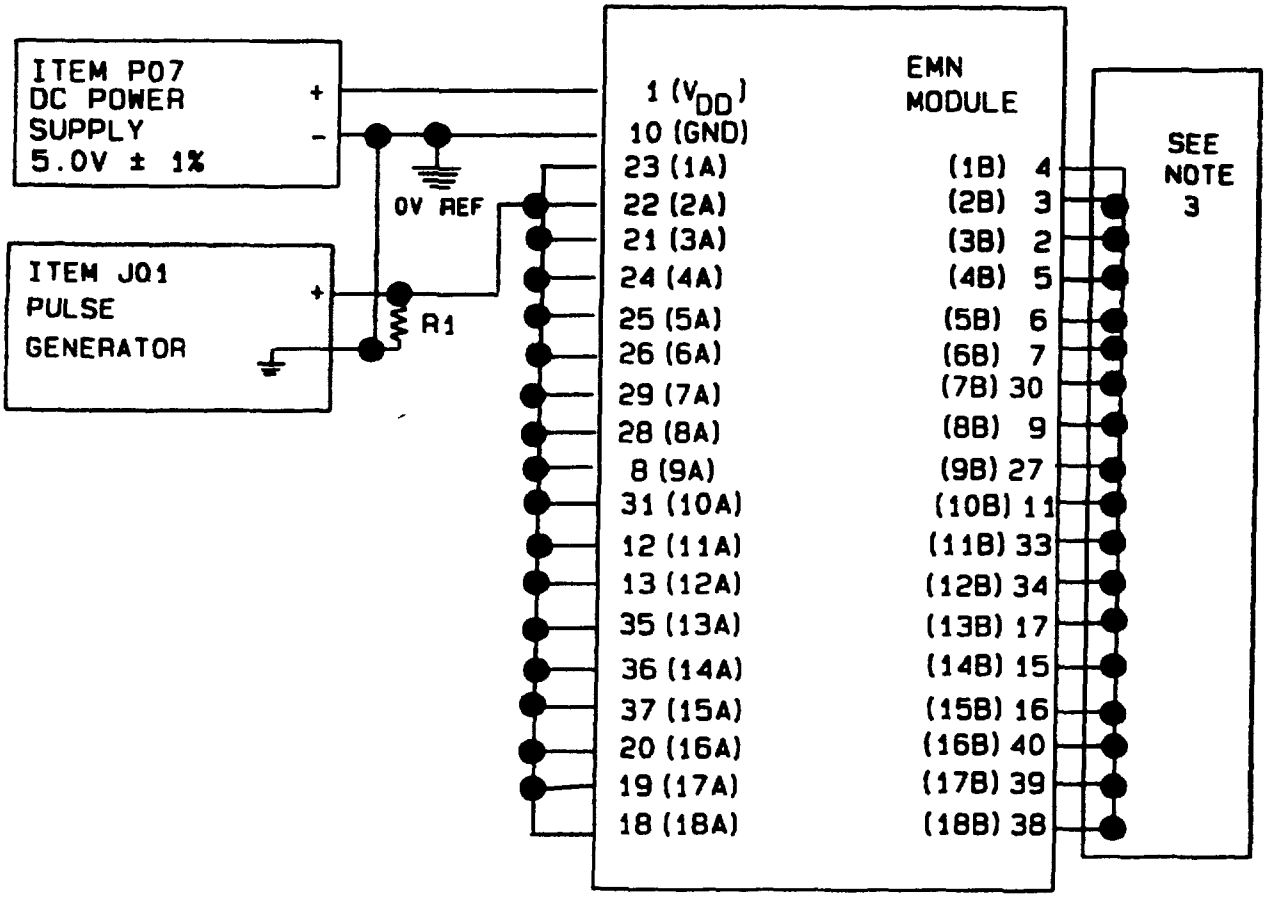
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.



NOTE: R1-R18 = 1.0 Ohm ± 1 per cent, 1/2 W.

* FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 per cent, 1/2 W.
2. Item J01 setup:
   PW = 200 ± 20 µS.
   PA = 5.0 ± 0.5 V.
   tTLH = tTHL = 50 ± 10 nS.
   Frequency = 50 kHz ± 20 per cent.
3. Load per figure 15.

* FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:11Z
Check the source to verify that this is the current version before use.

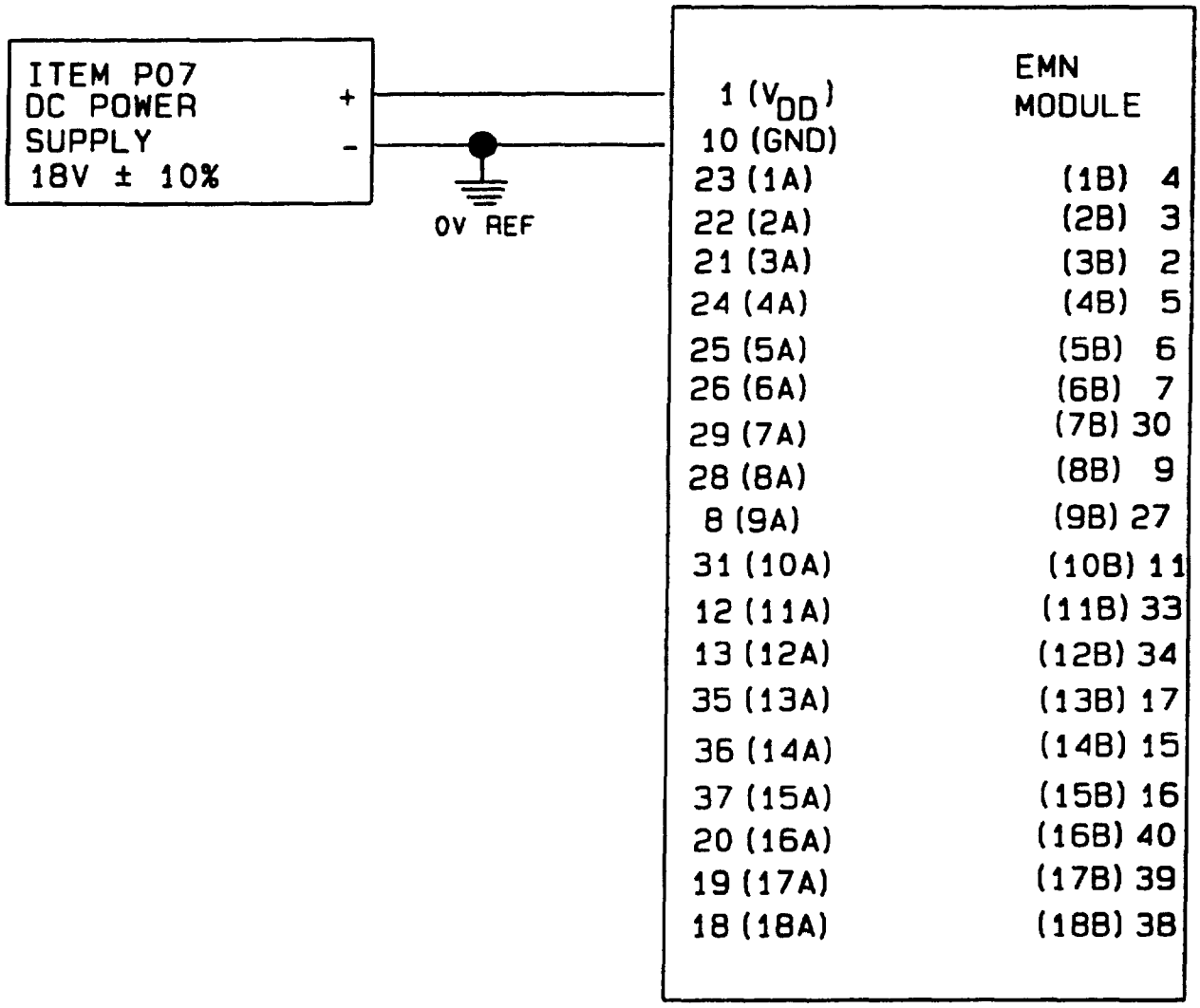


FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests C through U shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

*(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

* 6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:11Z
Check the source to verify that this is the current version before use.

* 6.3 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life parameters in table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	5.4	5.4	mW
Input current			
Low level			
(VIL = 0, VDD = 15V)	-45 maximum	-45 maximum	µA
High level			
(VIH = 15V, VDD = 15V)	200 maximum	200 maximum	nA
Output drive current (VDD = 5V)			
Low level (VOL = 0.4V)			
IOL = at -55°C	---	3.75 minimum	mA
0° and 25°C	3.2 minimum	3.2 minimum	mA
60°C	2.68 minimum	2.68 minimum	mA
85°C	---	2.1 minimum	mA
High level (VOH = 4.6V)			
IOH = at -55°	---	-1.85 minimum	mA
0° and 25°C	-1.25 minimum	-1.25 minimum	mA
60°C	-1.08 minimum	-1.08 minimum	mA
85°C	---	-0.9 minimum	mA

6.4 Module function. The module contains 18 independent inverting buffers which can be used as logic level shifters when VDD is the same as the VCC of the driven TTL/DTL circuit. The input signal high level (VIH) can exceed the VDD supply voltage when these devices are used for logic level conversion.

6.4.1 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$PTOTAL = (C\ VDD^2\ f)N + \text{Quiescent power dissipation}$$

Where: C = Load capacitance.
VDD = Supply voltage.
f = Operating frequency.
N = Number of outputs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

6.5 Module handling consideration. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

(a) Device pins should be in contact with conductive material except when being tested or in actual operation, to avoid build-up of static charge.

(b) Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

(c) Signals should not be applied to the inputs while the device power supply is off.

(d) All input signals and conditioning voltages must be removed before VDD is disconnected.

(e) Input voltages greater than the supply (VDD - VSS) should not be applied as this will forward bias the protection input diodes.

(f) All unused input leads must be connected to either VSS (GND) or VDD (device supply), whichever is appropriate for the logic circuit involved.

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

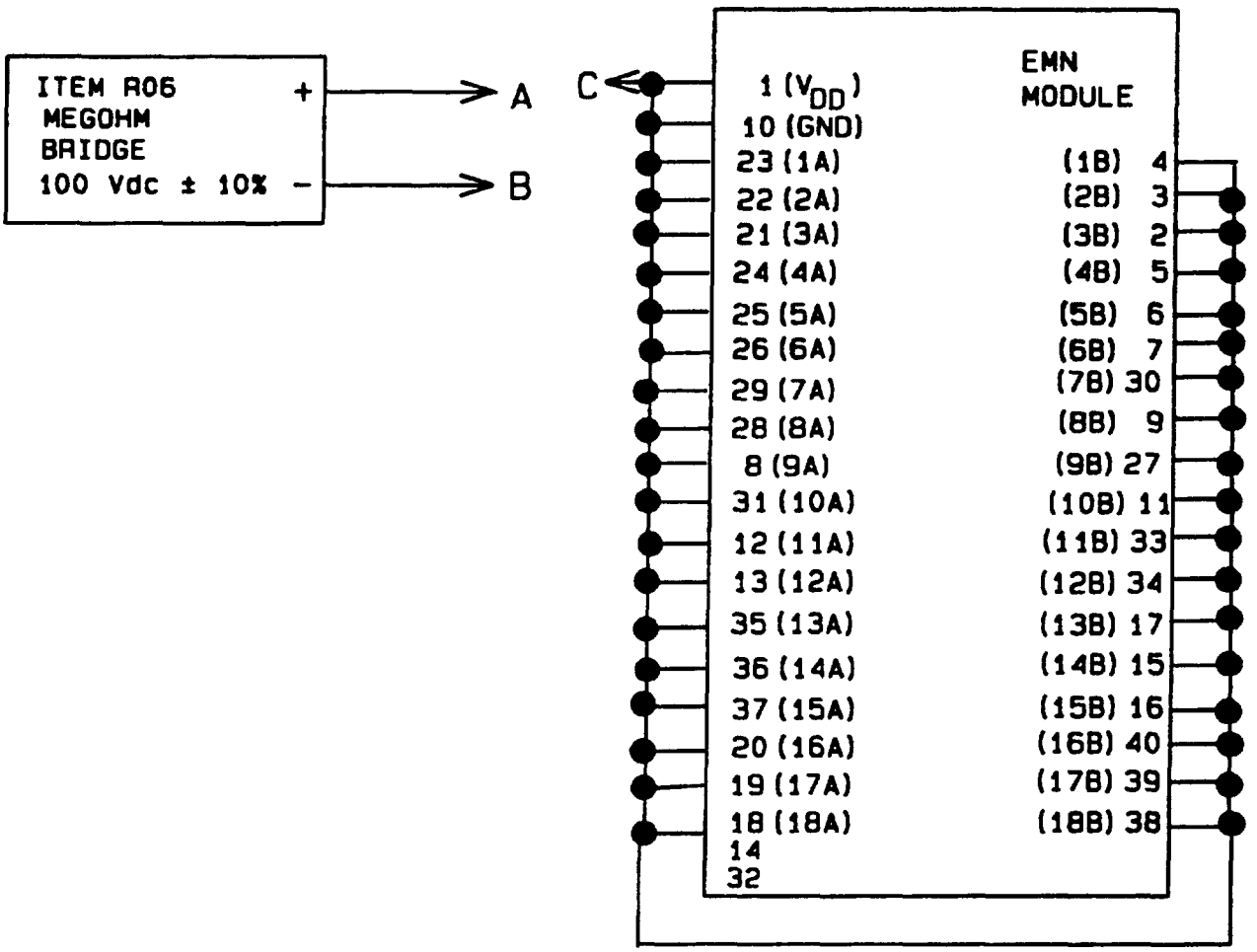
(Project 5963-0291)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test cond	Parameter	Connect		Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
A1	Isolation	C, 14, 32	Frame	10	INF	MOhm	5
A2	Isolation	14, 32	C	10	INF	MOhm	5

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:11Z
Check the source to verify that this is the current version before use.



WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

NOTE: When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

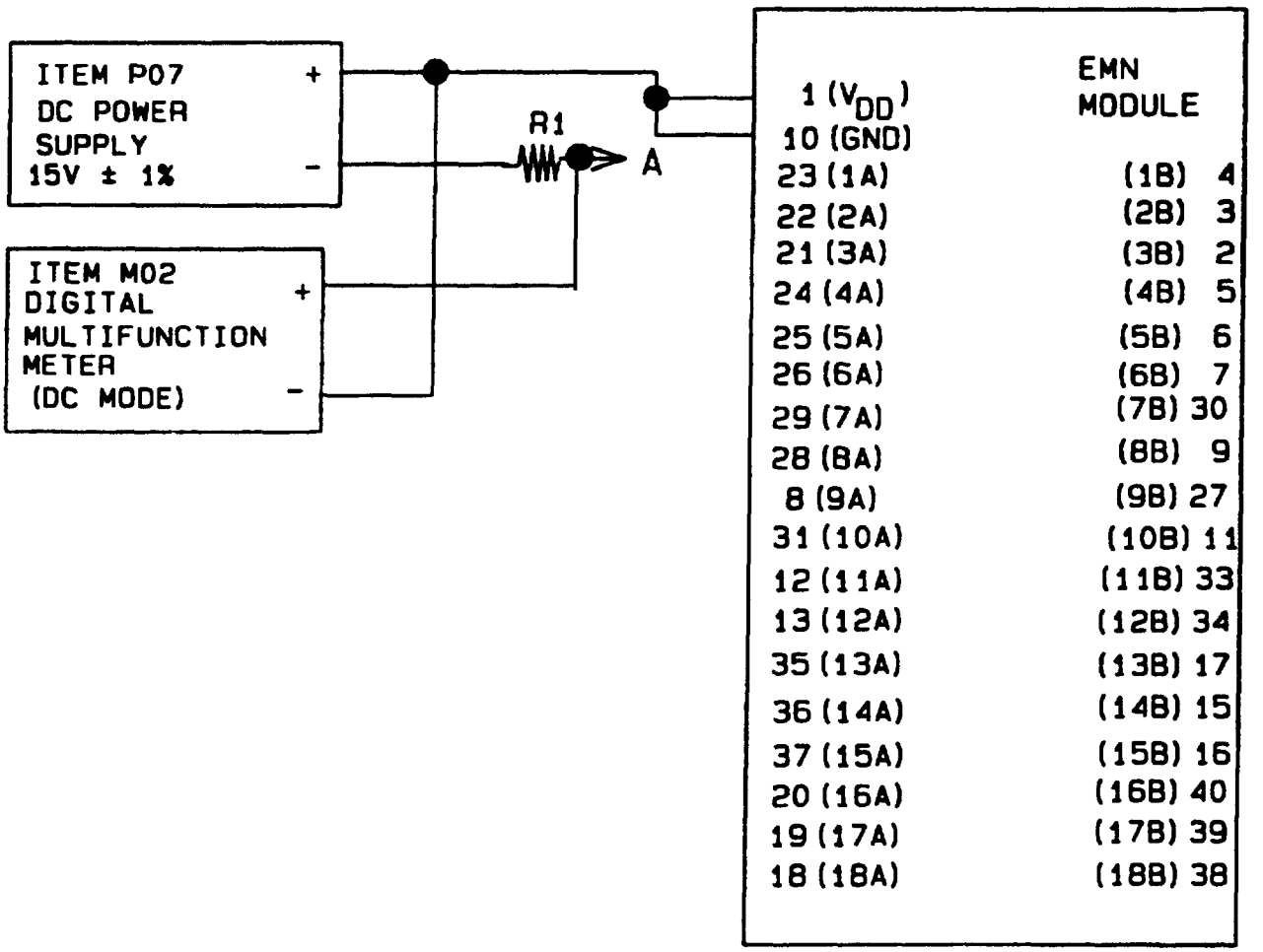
* **FIGURE 5. Isolation test circuit for table II.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE III. Input clamp diode voltage (negative) test.

Test cond	Parameter	Connect	Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	Min	Max		
B1	1A Input clamp diode voltage, negative	23	-0.4	-1.5	V	6
B2	2A	22				
B3	3A	21				
B4	4A	24				
B5	5A	25				
B6	6A	26				
B7	7A	29				
B8	8A	28				
B9	9A	8				
B10	10A	31				
B11	11A	12				
B12	12A	13				
B13	13A	35				
B14	14A	36				
B15	15A	37				
B16	16A	20				
B17	17A	19				
B18	18A Input clamp diode voltage, negative	18	-0.4	-1.5	V	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 13.7 kOhms ± 1 per cent, 1/8 W.

FIGURE 6. Input clamp diode voltage (negative) test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE IV. Output voltage (low-level) test.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25 ± 5°C ambient: Min	Initial limits 25 ± 5°C ambient: Max	Unit	Test fig
C1	1B Output voltage, low level, VDD = 5V	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	Open	4	0.0	0.4	V	7
C2	2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22		3				
C3	3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21		2				
C4	4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24		5				
C5	5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25		6				
C6	6B	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26		7				
C7	7B	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29		30				
C8	8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28		9				
C9	9B Output voltage, low level, VDD = 5V	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8	Open	27	0.0	0.4	V	7

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE IV. Output voltage (low-level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
C10	10B Output voltage, low level, VDD = 5V	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	Open	11	0.0	0.4	V	7
C11	11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12		33				
C12	12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13		34				
C13	13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35		17				
C14	14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36		15				
C15	15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37		16				
C16	16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20		40				
C17	17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19		39				
C18	18B Output voltage, low level, VDD = 5V	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	Open	38	0.0	0.4	V	7

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T17:11Z
Check the source to verify that this is the currentversion before use.

TABLE V. Output voltage (high-level) test.

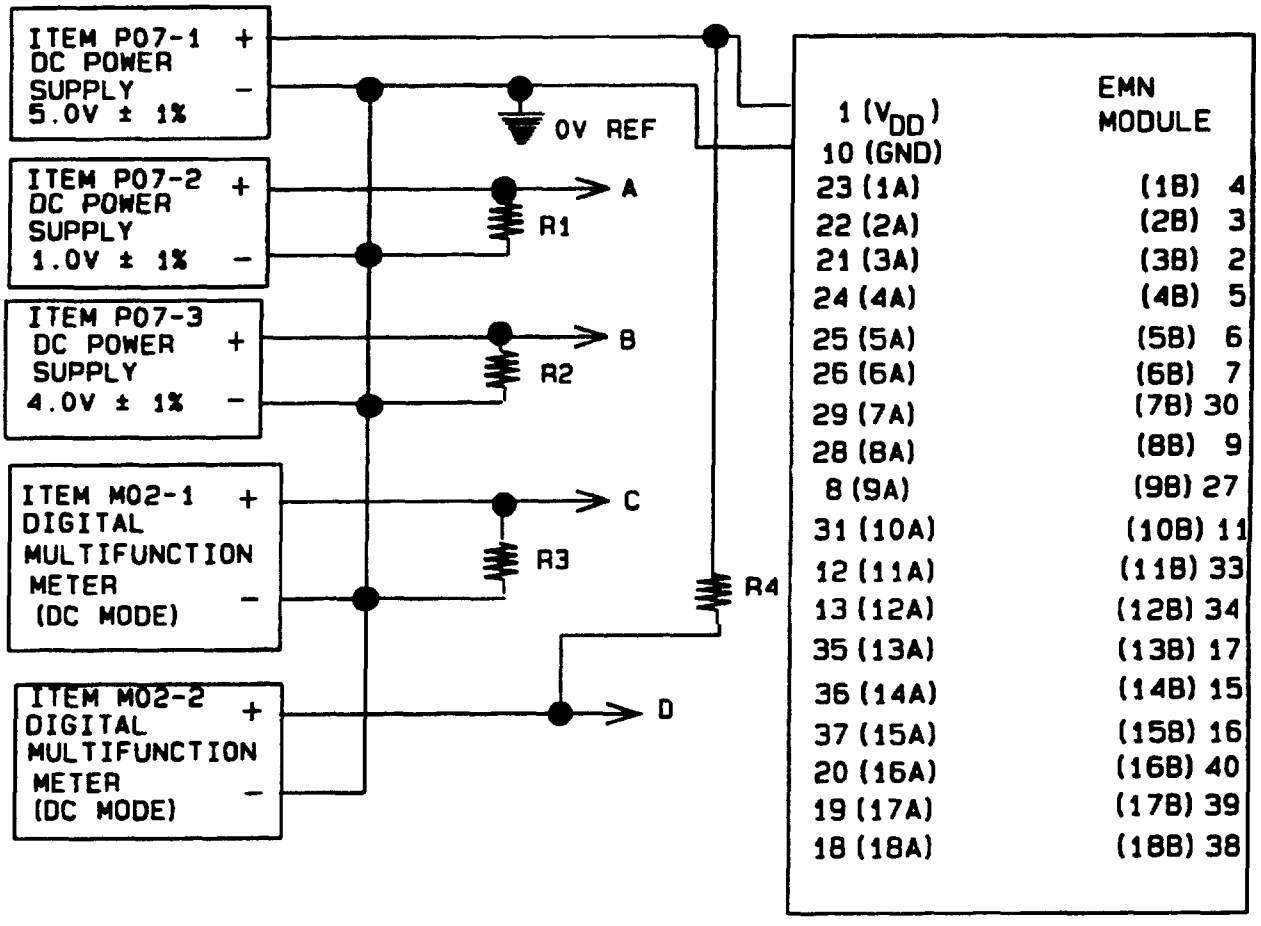
Test cond	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	1B Output voltage, high level, VDD = 5V	23	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	4	Open	2.5	5.0	V	7
D2	2B	22	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	3					
D3	3B	21	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	2					
D4	4B	24	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	5					
D5	5B	25	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	6					
D6	6B	26	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	7					
D7	7B	29	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	30					
D8	8B	28	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	9					
D9	9B	8	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	27					
D10	10B Output voltage, high level, VDD = 5V	31	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	11	Open	2.5	5.0	V	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE V. Output voltage (high-level) test - Continued

Test cond	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D11	11B Output voltage, high level, VDD = 5V	12	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	33	Open	2.5	5.0	V	7
D12	12B	13	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	34					
D13	13B	35	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	17					
D14	14B	36	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	15					
D15	15B	37	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	16					
D16	16B	20	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	40					
D17	17B	19	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	39					
D18	18B Output voltage, high level, VDD = 5V	18	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	38	Open	2.5	5.0	V	7

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 per cent, 1/2 W.
2. R3 = at 0° and 25°C - 2.0 kOhms ± 1 per cent, 1/8 W.
   at 60°C - 2.32 kOhms ± 1 per cent, 1/8 W.
   at 85°C - 2.8 kOhms ± 1 per cent, 1/8 W.
   at -55°C - 1.37 kOhms ± 1 per cent, 1/8 W.
3. R4 = at 0° and 25°C - 1.43 kOhms ± 1 per cent, 1/8 W.
   at 60°C - 1.69 kOhms ± 1 per cent, 1/8 W.
   at 85°C - 2.21 kOhms ± 1 per cent, 1/8 W.
   at -55°C - 1.24 kOhms ± 1 per cent, 1/8 W.

* FIGURE 7. Output voltage (low-level and high-level) test circuit for tables IV and V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (low-level) test.

Test cond	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1B Output voltage, low level, VDD = 15V	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	Open	4	Open	0.0	1.5	V	8
E2	2B			3					
E3	3B			2					
E4	4B			5					
E5	5B			6					
E6	6B			7					
E7	7B			30					
E8	8B			9					
E9	9B			27					
E10	10B Output voltage, low level, VDD = 15V	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	Open	11	Open	0.0	1.5	V	8

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:11Z
Check the source to verify that this is the currentversion before use.

TABLE VI. Output voltage (low-level) test - Continued.

Test cond	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E11	11B Output voltage, low level, VDD = 15V	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	Open	33	Open	0.0	1.5	V	8
E12	12B			34					
E13	13B			17					
E14	14B			15					
E15	15B			16					
E16	16B			40					
E17	17B			39					
E18	18B Output voltage, low level, VDD = 15V	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	Open	38	Open	0.0	1.5	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (high-level) test.

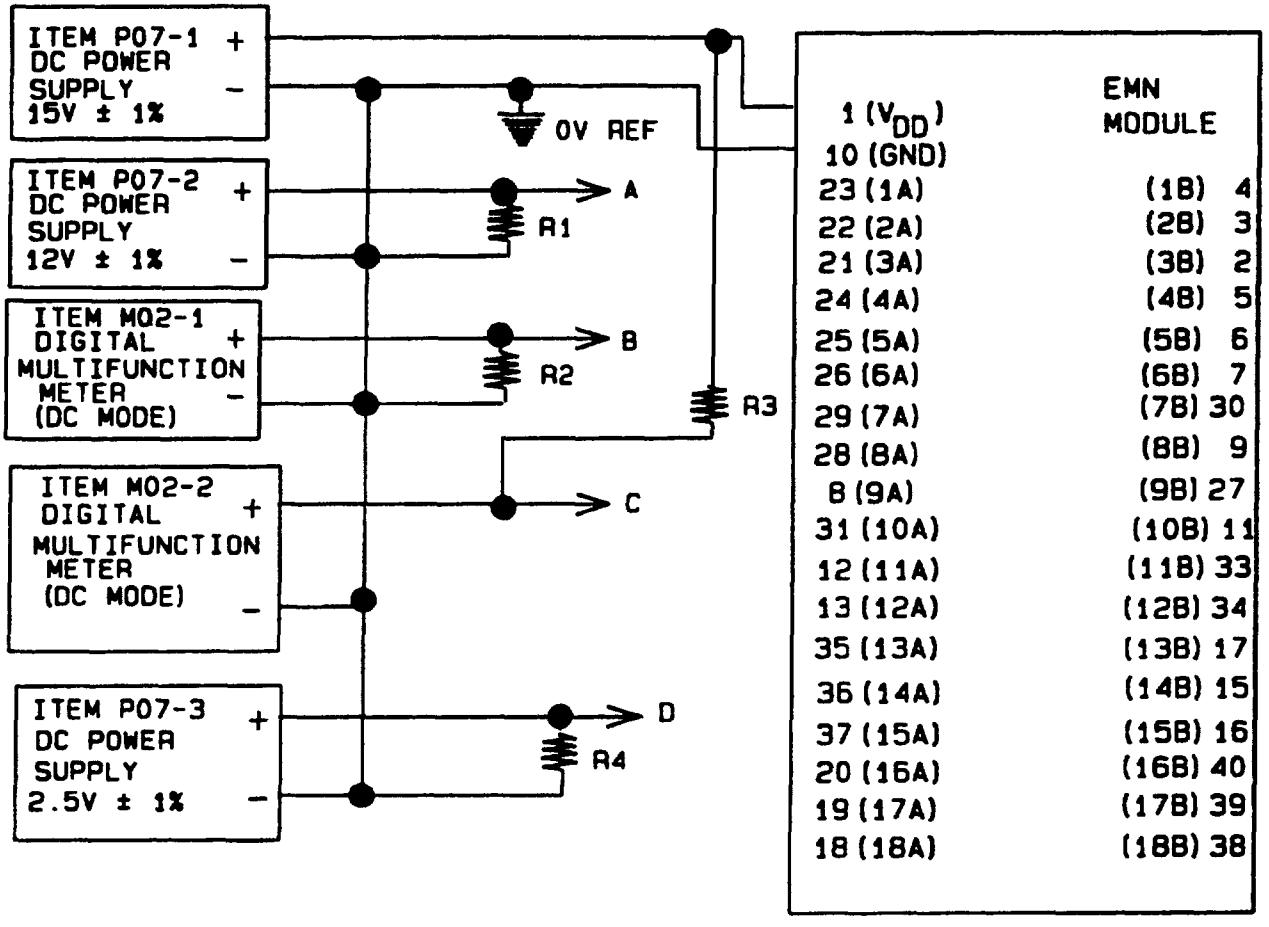
Test cond	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1B Output voltage, high level, VDD = 15V	Open	4	Open	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	13.5	15.15	V	8
F2	2B		3						
F3	3B		2						
F4	4B		5						
F5	5B		6						
F6	6B		7						
F7	7B		30						
F8	8B Output voltage, high level, VDD = 15V	Open	9	Open	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	13.5	15.15	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (high-level) test -Continued.

Test cond	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F9	9B Output voltage, high level, VDD = 15V	Open	27	Open	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	13.5	15.15	V	8
F10	10B		11						
F11	11B		33						
F12	12B		34						
F13	13B		17						
F14	14B		15						
F15	15B		16						
F16	16B		40						
F17	17B		39						
F18	18B Output voltage, high level, VDD = 15V	Open	38	Open	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	13.5	15.15	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 596 Ohms ± 1 per cent, 1/4 W.
2. R2 = at 0° and 25°C - 3.6 kOhms ± 1 per cent, 1/8 W.
   at 60°C - 4.22 kOhms ± 1 per cent, 1/8 W.
   at 85°C - 4.99 kOhms ± 1 per cent, 1/8 W.
   at -55°C - 2.49 kOhms ± 1 per cent, 1/8 W.
3. R3 = at 0° and 25°C - 562 kOhms ± 1 per cent, 1/8 W.
   at 60°C - 665 Ohms ± 1 per cent, 1/8 W.
   at 85°C - 806 Ohms ± 1 per cent, 1/8 W.
   at -55°C - 453 Ohms ± 1 per cent, 1/8 W.
4. R4 = 51.1 Ohms ± 5 per cent, 1/2 W.

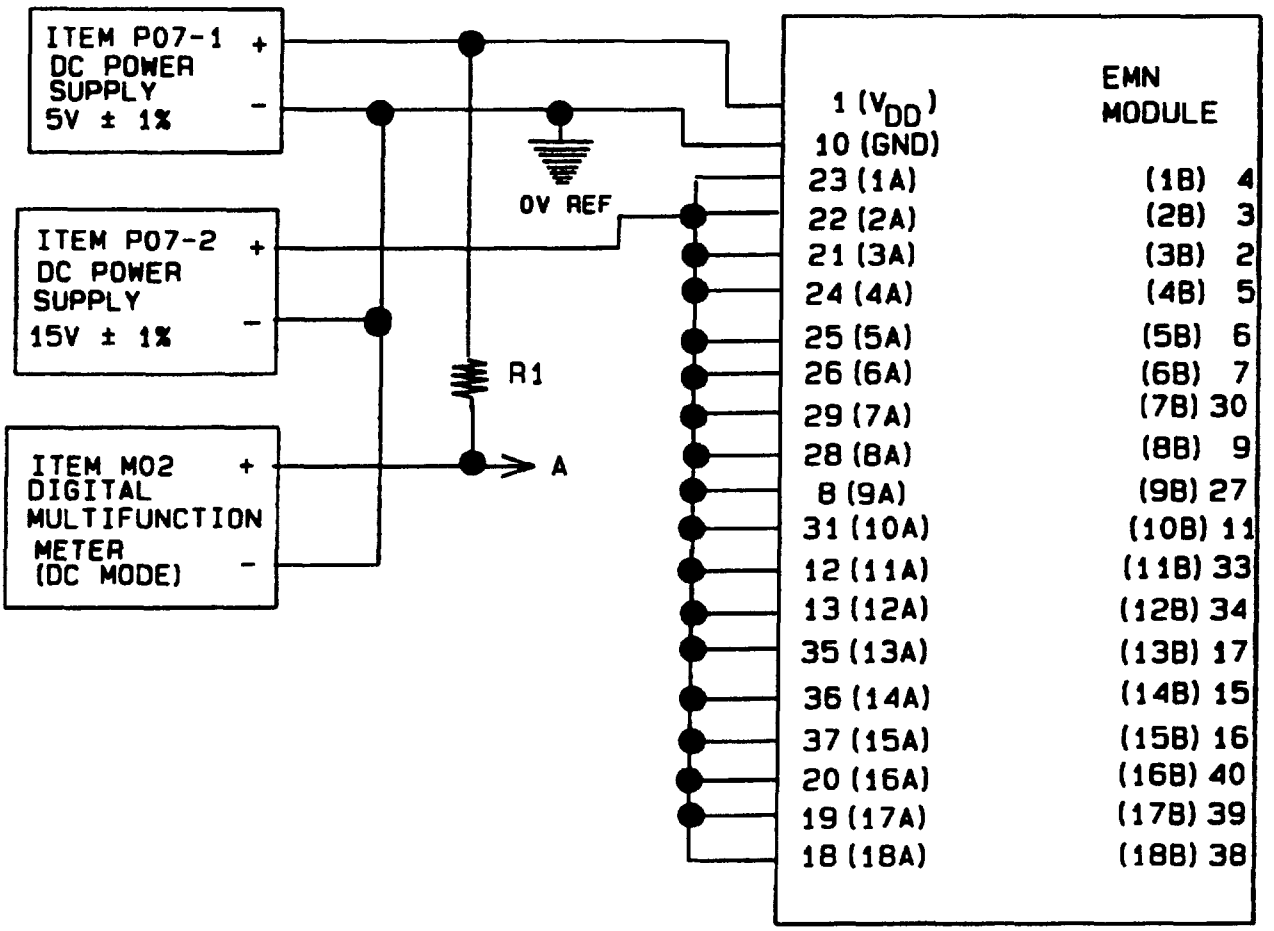
* FIGURE 8. Output voltage (low-level and high-level) test circuit for tables VI and VII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE VIII. Logic level conversion test.

Test cond	Parameter	Connect A to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
G1	1B Logic level conversion	4	0.0	0.4	V	9
G2	2B	3				
G3	3B	2				
G4	4B	5				
G5	5B	6				
G6	6B	7				
G7	7B	30				
G8	8B	9				
G9	9B	27				
G10	10B	11				
G11	11B	33				
G12	12B	34				
G13	13B	17				
G14	14B	15				
G15	15B	16				
G16	16B	40				
G17	17B	39				
G18	18B Logic level conversion	38	0.0	0.4	V	9

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:11Z
Check the source to verify that this is the current version before use.



NOTE: R1 = at 0° and 25°C - 1.43 kOhms ± 1 per cent, 1/8 W.
at 60°C - 1.69 kOhms ± 1 per cent, 1/8 W.
at 85°C - 2.21 kOhms ± 1 per cent, 1/8 W.
at -55°C - 1.24 kOhms ± 1 per cent, 1/8 W.

* FIGURE 9. Logic level conversion test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test.

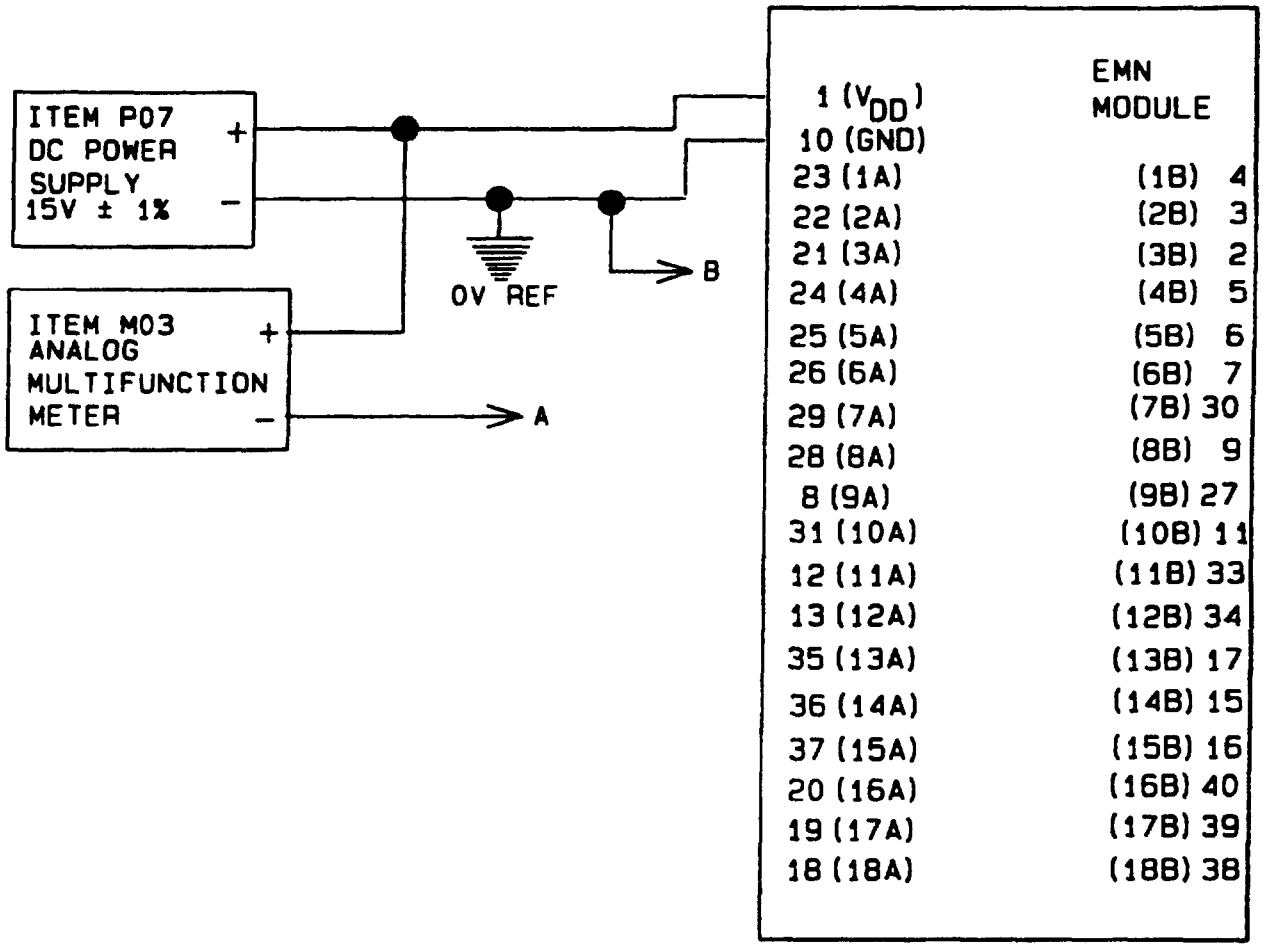
Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
H1	1A Input current, high level	23	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	0	200	nA	10
H2	2A	22	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18				
H3	3A	21	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18				
H4	4A	24	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18				
H5	5A	25	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18				
H6	6A	26	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18				
H7	7A	29	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18				
H8	8A	28	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18				
H9	9A	8	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18				
H10	10A	31	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18				
H11	11A Input current, high level	12	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	0	200	nA	10

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
H12	12A Input current, high level	13	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	0	200	nA	10
H13	13A	35	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18				
H14	14A	36	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18				
H15	15A	37	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18				
H16	16A	20	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18				
H17	17A	19	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18				
H18	18A Input current, high level	18	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	0	200	nA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.



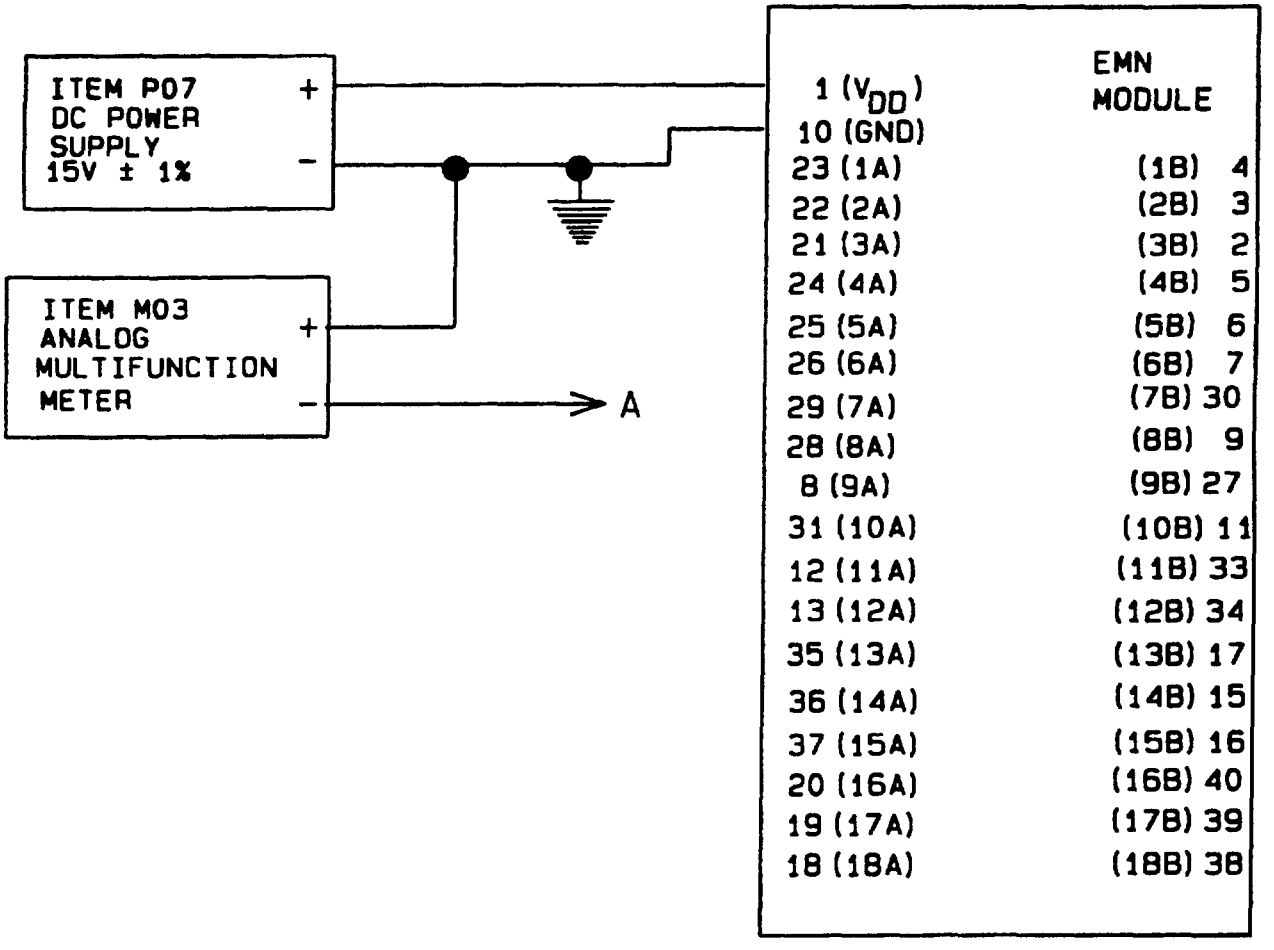
* FIGURE 10. Input current (high-level) test circuit for table .

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE X. Input current (low-level) test.

Test cond	Parameter	Connect A to pin	Initial limits 25 ± 5°C ambient		Unit	Test fig
			Min	Max		
J1	1A Input current, low level	23	-30	-45	μA	11
J2	2A	22				
J3	3A	21				
J4	4A	24				
J5	5A	25				
J6	6A	26				
J7	7A	29				
J8	8A	28				
J9	9A	8				
J10	10A	31				
J11	11A	12				
J12	12A	13				
J13	13A	35				
J14	14A	36				
J15	15A	37				
J16	16A	20				
J17	17A	19				
J18	18A Input current, low level	18	-30	-45	μA	11

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:11Z
Check the source to verify that this is the current version before use.



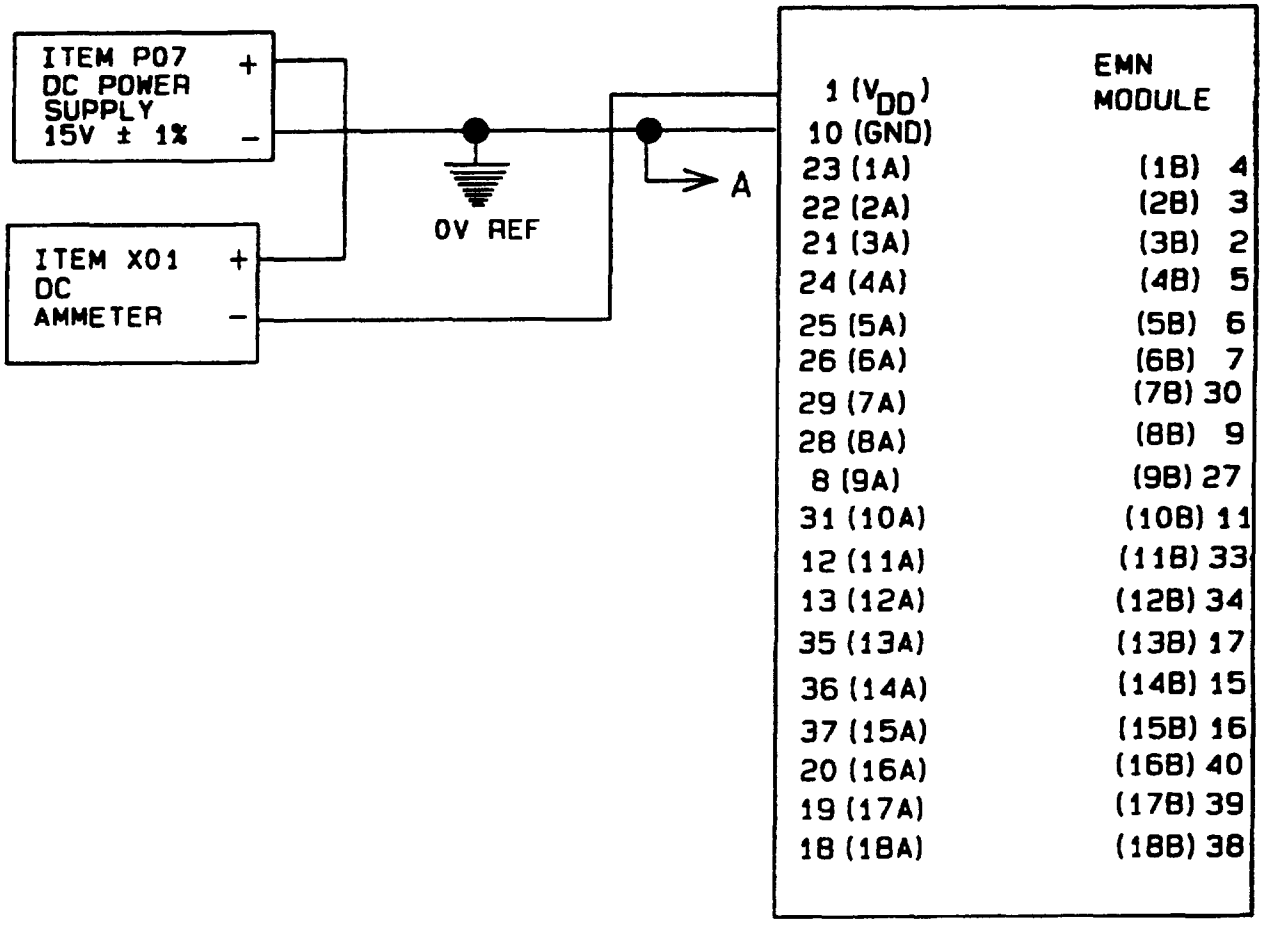
* FIGURE 11. Input current (low-level) test circuit for table

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

*TABLE XI. Power supply current (low-level and high level) test.

Test cond	Parameter	Connect A to pin	Initial limits 25 ± 5°C ambient Min	Max	Unit	Test fig
K1	Power supply current, low level	Open	0	12	μA	12
K2	Power supply current, high level	23,22,21,24, 25,26,29,28, 8,31,12,13, 35,36,37,20, 19,18	450	687	μA	12

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:11Z
Check the source to verify that this is the current version before use.



CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 12. Power supply current (low-level and high-level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XII. Transition time (tTLH) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
L1	1A to 1B Transition time (tTLH), CL = 50 pF	22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	23	4	15	200	nS	13,14
L2	2A to 2B	23,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	22	3				
L3	3A to 3B	23,22,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	21	2				
L4	4A to 4B	23,22,21,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	24	5				
L5	5A to 5B	23,22,21,24,26,29,28,8, 31,12,13,35,36,37,20,19,18	25	6				
L6	6A to 6B	23,22,21,24,25,29,28,8, 31,12,13,35,36,37,20,19,18	26	7				
L7	7A to 7B	23,22,21,24,25,26,28,8, 31,12,13,35,36,37,20,19,18	29	30				
L8	8A to 8B	23,22,21,24,25,26,29,8, 31,12,13,35,36,37,20,19,18	28	9				
L9	9A to 9B	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8	27				
L10	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
L11	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
L12	12A to 12B Transition time (tTLH), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34	15	200	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XII. Transition time (tTLH) test - Continued.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
L13	13A to 13B Transition time (tTLH), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17	15	200	nS	13,14
L14	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
L15	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
L16	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
L17	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				
L18	18A to 18B Transition time (tTLH), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	15	200		
L19	1A to 1B Transition time (tTLH), CL = 150 pF	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,29,18	23	4	35	400		
L20	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
L21	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
L22	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
L23	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
L24	6A to 6B Transition time (tTLH), CL = 150 pF	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7	35	400	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XII. Transition time (tTLH) test - Continued.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
L25	7A to 7B Transition time (tTLH), CL = 150 pF	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30	35	400	nS	13,14
L26	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
L27	9A to 9B	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8	27				
L28	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
L29	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
L30	12A to 12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
L31	13A to 13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
L32	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
L33	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
L34	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
L35	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				
L36	18A to 18B Transition time (tTLH), CL = 150 pF	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	35	400	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Transition time (tTHL) test.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
M1	1A to 1B Transition time (tTHL), CL = 50 pF	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	5	80	nS	13,14
M2	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
M3	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
M4	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
M5	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
M6	6A to 6B	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
M7	7A to 7B	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
M8	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
M9	9A to 9B	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8	27				
M10	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
M11	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
M12	12A to 12B Transition time (tTHL), CL = 50 pF	23,22,21,24,25,26,29,28,8, 12,35,36,37,20,19,18	13	34	5	80	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Transition time (tTHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
M13	13A to 13B Transition time (tTHL), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17	5	80	nS	13,14
M14	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
M15	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
M16	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
M17	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				
M18	18A to 18B Transition time (tTHL), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	5	80		
M19	1A to 1B Transition time (tTHL), CL = 150 pF	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	15	140		
M20	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
M21	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
M22	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
M23	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
M24	6A to 6B Transition time (tTHL), CL = 150 pF	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7	15	140	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XIII. Transition time (tTHL) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
M25	7A to 7B Transition time (tTHL), CL = 150 pF	23,22,21,24,25,26,28,8,31,12,13,35,36,37,20,19,18	29	30	15	140	nS	13,14
M26	8A to 8B	23,22,21,24,25,26,29,8,31,12,13,35,36,37,20,19,18	28	9				
M27	9A to 9B	23,22,21,24,25,26,29,28,31,12,13,35,36,37,20,19,18	8	27				
M28	10A to 10B	23,22,21,24,25,26,29,28,8,12,13,35,36,37,20,19,18	31	11				
M29	11A to 11B	23,22,21,24,25,26,29,28,8,31,13,35,36,37,20,19,18	12	33				
M30	12A to 12B	23,22,21,24,25,26,29,28,8,31,12,35,36,37,20,19,18	13	34				
M31	13A to 13B	23,22,21,24,25,26,29,28,8,31,12,13,36,37,20,19,18	35	17				
M32	14A to 14B	23,22,21,24,25,26,29,28,8,31,12,13,35,37,20,19,18	36	15				
M33	15A to 15B	23,22,21,24,25,26,29,28,8,31,12,13,35,36,20,19,18	37	16				
M34	16A to 16B	23,22,21,24,25,26,29,28,8,31,12,13,35,36,37,19,18	20	40				
M35	17A to 17B	23,22,21,24,25,26,29,28,8,31,12,13,35,36,37,20,18	19	39				
M36	18A to 18B Transition time (tTHL), CL = 150 pF	23,22,21,24,25,26,29,28,8,31,12,13,35,36,37,20,19	18	38	15	140	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPLH) test.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
N1	1A to 1B Propagation delay time (tPLH), CL = 50 pF	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	10	160	nS	13,14
N2	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
N3	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
N4	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
N5	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
N6	6A to 6B	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
N7	7A to 7B	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
N8	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
N9	9A to 9B	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8	27				
N10	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
N11	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
N12	12A to 12B Propagation delay time (tPLH), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34	10	160	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPLH) test - Continued.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
N13	13A to 13B Propagation delay time (tPLH), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17	10	160	nS	13,14
N14	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
N15	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
N16	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
N17	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				
N18	18A to 18B Propagation delay time (tPLH), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	10	160		
N19	1A to 1B Propagation delay time (tPLH), CL = 150 pF	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	15	240		
N20	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
N21	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
N22	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
N23	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
N24	6A to 6B Propagation delay time (tPLH), CL = 150 pF	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7	15	240	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (tPLH) test - Continued.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
N25	7A to 7B Propagation delay time (tPLH), CL = 150 pF	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30	15	240	nS	13,14
N26	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
N27	9A to 9B	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8	27				
N28	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
N29	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
N30	12A to 12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
N31	13A to 13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
N32	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
N33	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
N34	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
N35	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				
N36	18A to 18B Propagation delay time (tPLH), CL = 150 pF	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	15	240	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (tPHL) test.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P1	1A to 1B Propagation delay time (tPHL), CL = 50 pF	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	5	60	nS	13,14
P2	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
P3	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
P4	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
P5	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
P6	6A to 6B	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7				
P7	7A to 7B	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30				
P8	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
P9	9A to 9B	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8	27				
P10	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
P11	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
P12	12A to 12B Propagation delay time (tPHL), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34	5	60	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (tPHL) test - Continued.

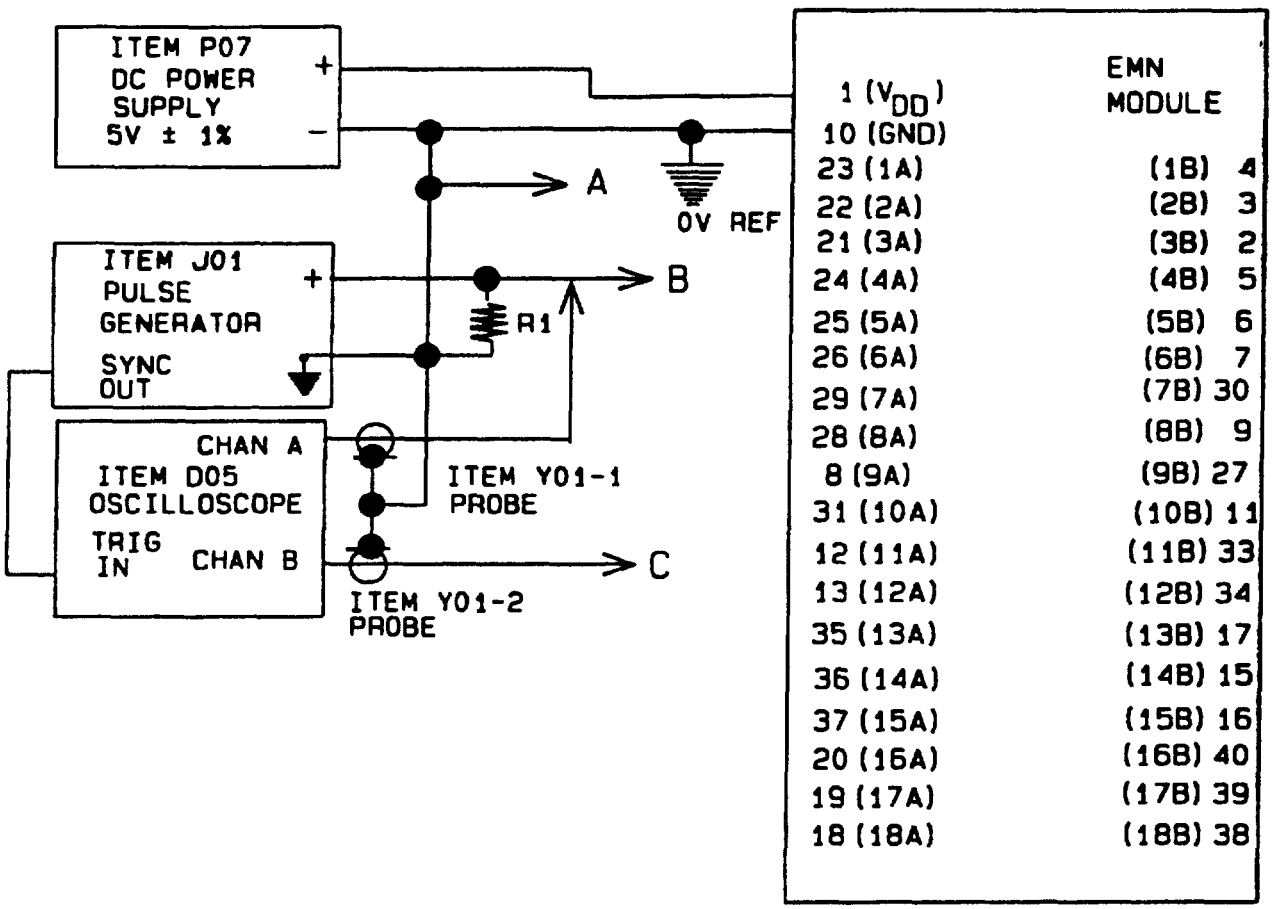
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
P13	13A to 13B Propagation delay time (tPHL), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17	5	60	nS	13,14
P14	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
P15	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
P16	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
P17	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				
P18	18A to 18B Propagation delay time (tPHL), CL = 50 pF	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	5	60		
P19	1A to 1B Propagation delay time (tPHL), CL = 150 pF	22,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	23	4	10	140		
P20	2A to 2B	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	22	3				
P21	3A to 3B	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	21	2				
P22	4A to 4B	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	24	5				
P23	5A to 5B	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	25	6				
P24	6A to 6B Propagation delay time (tPHL), CL = 150 pF	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	26	7	10	140	nS	13,14

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P25	7A to 7B Propagation delay time (tPHL), CL = 150 pF	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	29	30	10	140	nS	13,14
P26	8A to 8B	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	28	9				
P27	9A to 9B	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	8	27				
P28	10A to 10B	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	31	11				
P29	11A to 11B	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	12	33				
P30	12A to 12B	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	13	34				
P31	13A to 13B	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	35	17				
P32	14A to 14B	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	36	15				
P33	15A to 15B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,20,19,18	37	16				
P34	16A to 16B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	20	40				
P35	17A to 17B	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	19	39				
P36	18A to 18B Propagation delay time (tPHL), CL = 150 pF	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	18	38	10	140	nS	13,14

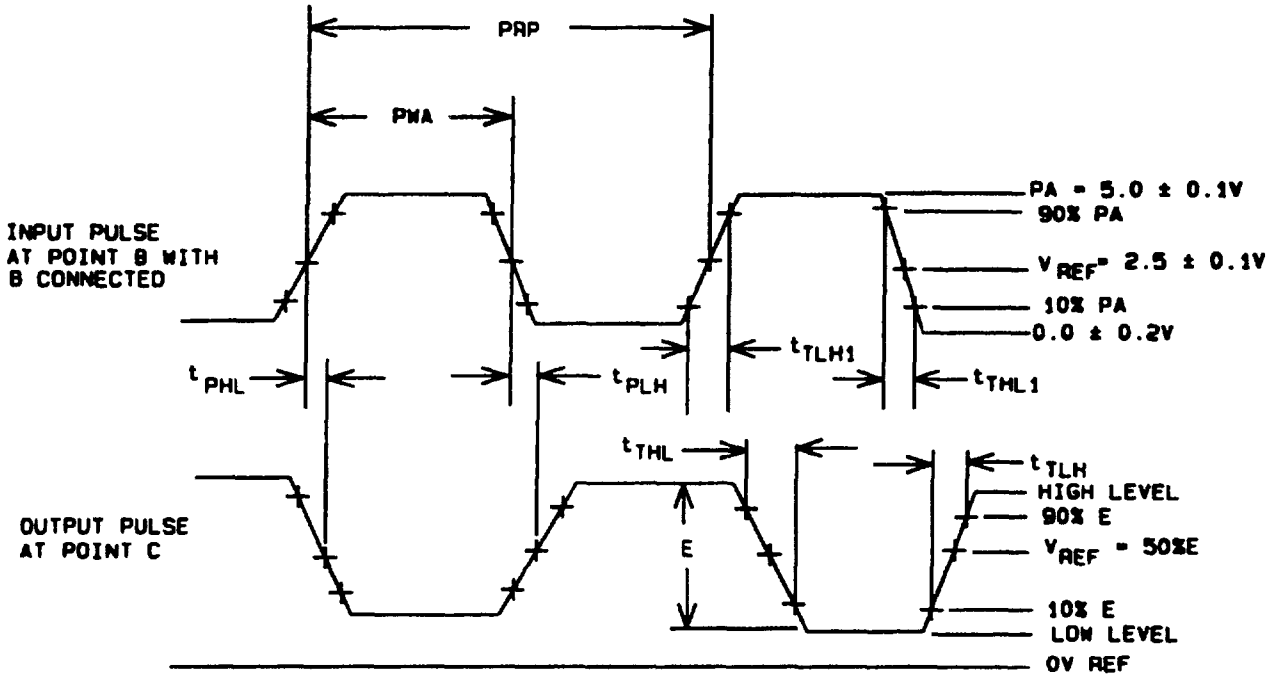
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51.1 Ohms ± 5 per cent, 1/2 W.
2. Load all outputs with load as specified on figure 15.

FIGURE 13. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit for tables XII through XV.

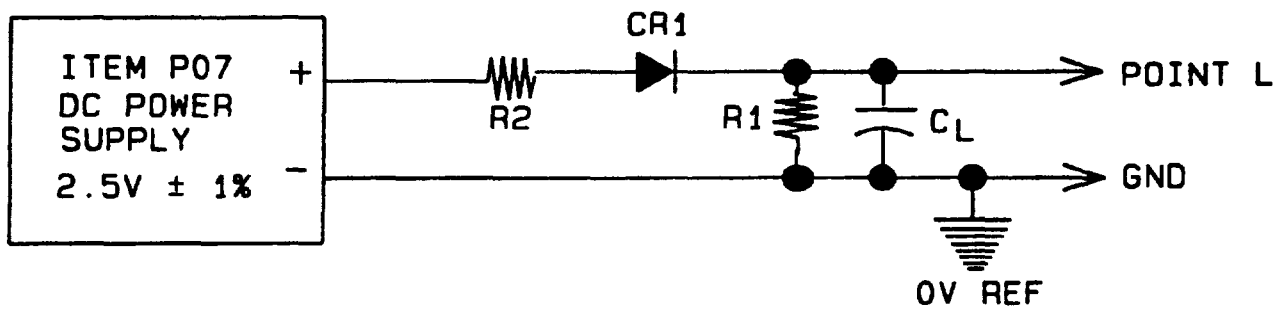
Source: https://assist.dla.mil -- Downloaded: 20261O-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters: PRP = 100 ± 10 µS.
   PWA = 50 ± 5 µS.
   tTLH1 = tTHL1 = 10 ± 2 nS.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.4 V.
3. The lead length of electrical conductors between resistor R1, and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 14. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit waveforms for tables XII through XV.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N4150 or equivalent.
2. R1 = 2.8 kOhms ± 1 per cent, 1/8 W.
3. R2 = Select a value for a current flow of 2.1 mA at point L with point L held at 0.4V.
4. CL = 50 ± 5 pF (including probe and parasitic capacitance) for test conditions L1-L18, M1-M18, N1-N18 and P1-P18.
   = 150 ± 10 pF (including probe and parasitic capacitance) for test conditions L19-L36, M19-M36, N19-N36 and P19-P36.
5. The maximum lead length from point L to the appropriate module pin shall be less than 1 inch (25.4 mm).

FIGURE 15. EMN logic load circuit, inverting buffer.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk (low level) test.

Test cond	Parameter	Connect				Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to open	C to pin	D to pin	Min	Max		
Q1	1B Output crosstalk, low level	23	Open	22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	4	0	400	mV	16,17
Q2	2B	22		23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	3				
Q3	3B	21		23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	2				
Q4	4B	24		23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	5				
Q5	5B	25		23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	6				
Q6	6B	26		23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	7				
Q7	7B	29		23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	30				
Q8	8B	28		23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	9				
Q9	9B	8		23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	27				
Q10	10B	31		23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	11				
Q11	11B	12		23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	33				
Q12	12B Output crosstalk, low level	13	Open	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	34	0	400	mV	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk (low level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to open	Connect C to pin	Connect D to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
Q13	13B Output crosstalk, low level	35	Open	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	17	0	400	mV	16,17
Q14	14B	36		23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	15				
Q15	15B	37		23,22,21,24,25,26,29,28,8 31,12,13,35,36,20,19,18	16				
Q16	16B	20		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	40				
Q17	17B	19		23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	39				
Q18	18B Output crosstalk, low level	18	Open	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	38	0	400	mV	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XVII Output crosstalk (high level) test.

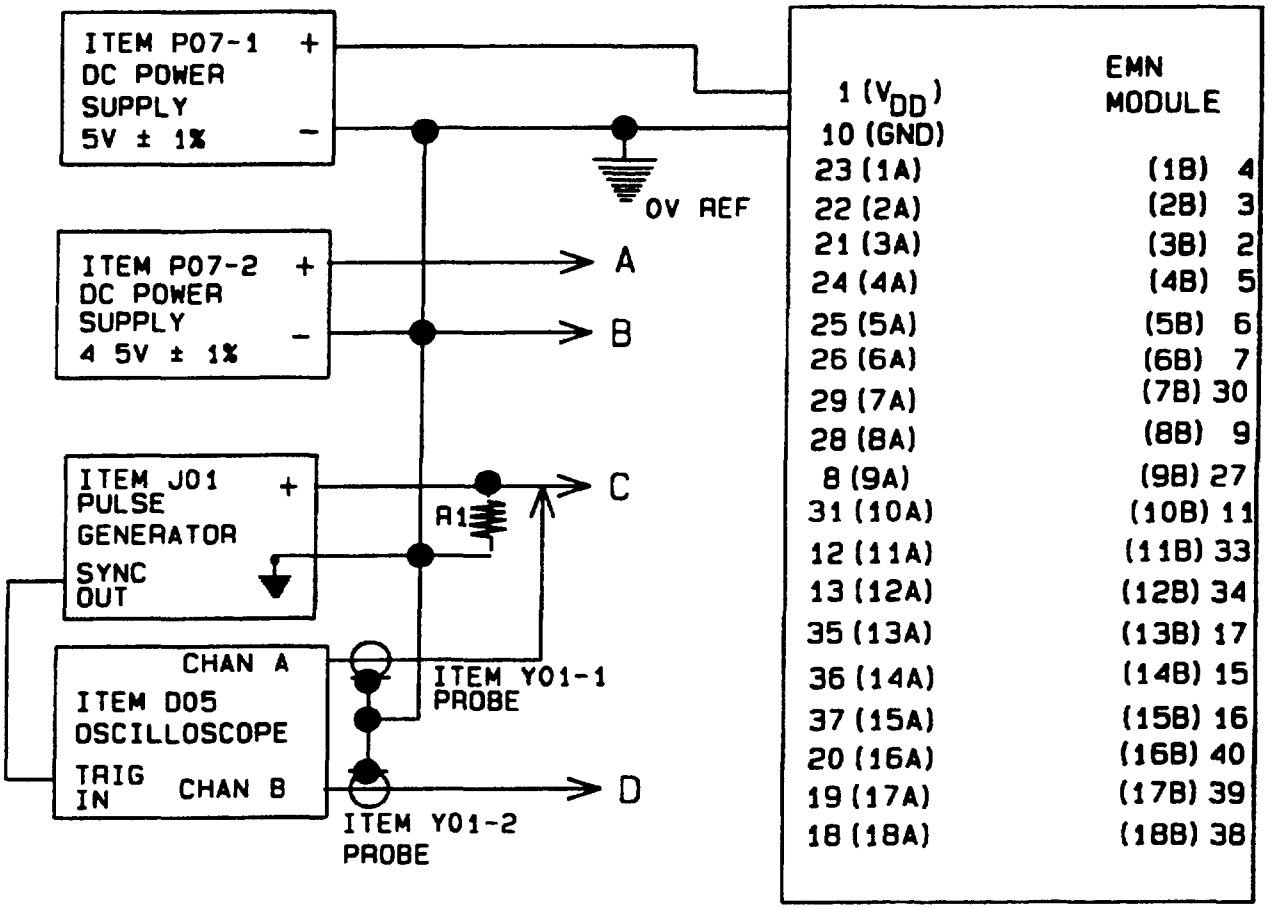
Test cond	Parameter	Connect: A to pin	Connect: B to open	Connect: C to pin	Connect: D to pin	Initial limits 25 ± 5°C ambient: Min	Initial limits 25 ± 5°C ambient: Max	Unit	Test fig
R1	1B Output crosstalk, high level	Open	23	22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19,18	4	0	400	mV	16,17
R2	2B		22	23,21,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	3				
R3	3B		21	23,22,24,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	2				
R4	4B		24	23,22,21,25,26,29,28,8,31, 12,13,35,36,37,20,19,18	5				
R5	5B		25	23,22,21,24,26,29,28,8,31, 12,13,35,36,37,20,19,18	6				
R6	6B		26	23,22,21,24,25,29,28,8,31, 12,13,35,36,37,20,19,18	7				
R7	7B		29	23,22,21,24,25,26,28,8,31, 12,13,35,36,37,20,19,18	30				
R8	8B		28	23,22,21,24,25,26,29,8,31, 12,13,35,36,37,20,19,18	9				
R9	9B		8	23,22,21,24,25,26,29,28, 31,12,13,35,36,37,20,19,18	27				
R10	10B		31	23,22,21,24,25,26,29,28,8, 12,13,35,36,37,20,19,18	11				
R11	11B		12	23,22,21,24,25,26,29,28,8, 31,13,35,36,37,20,19,18	33				
R12	12B Output crosstalk, high level	Open	13	23,22,21,24,25,26,29,28,8, 31,12,35,36,37,20,19,18	34	0	400	mV	16,17

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (high level) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to open	Connect: C to pin	Connect: D to pin	Initial limits 25 ± 5°C ambient: Min	Initial limits 25 ± 5°C ambient: Max	Unit	Test fig
R13	13B Output crosstalk, high level	Open	35	23,22,21,24,25,26,29,28,8, 31,12,13,36,37,20,19,18	17	0	400	mV	16,17
R14	14B		36	23,22,21,24,25,26,29,28,8, 31,12,13,35,37,20,19,18	15				
R15	15B		37	23,22,21,24,25,26,29,28,8 31,12,13,35,36,20,19,18	16				
R16	16B		20	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,19,18	40				
R17	17B		19	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,18	39				
R18	18B Output crosstalk, high level	Open	18	23,22,21,24,25,26,29,28,8, 31,12,13,35,36,37,20,19	38	0	400	mV	16,17

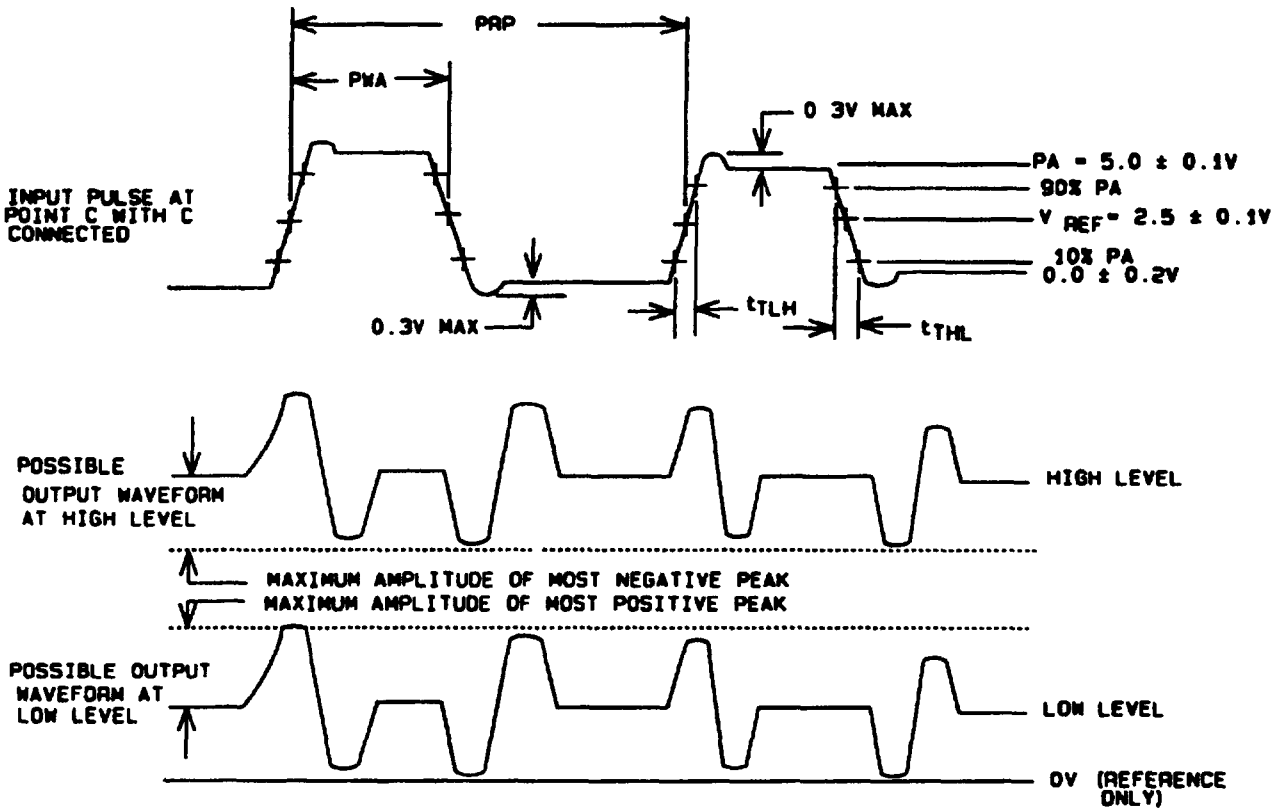
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51.1 Ohms ± 1 per cent, 1/2 W.
2. Load all outputs with load as specified on figure 15.

FIGURE 16. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters: (verify the input pulse parameters at each input pin before making each measurement).
   PRP = 50 ± 5 µS.
   PWA = 25 ± 2.5 µS.
   tTLH = tTHL = 20 ± 2 nS.
2. The lead length of electrical conductors between resistor R1 and input pins, probes Y01-1, Y01-2 to input and output pins, and ground leads of probes Y01-1, Y01-2 to test fixture and module ground pins shall be less than 1 inch (25.4 mm).
3. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.4 V.

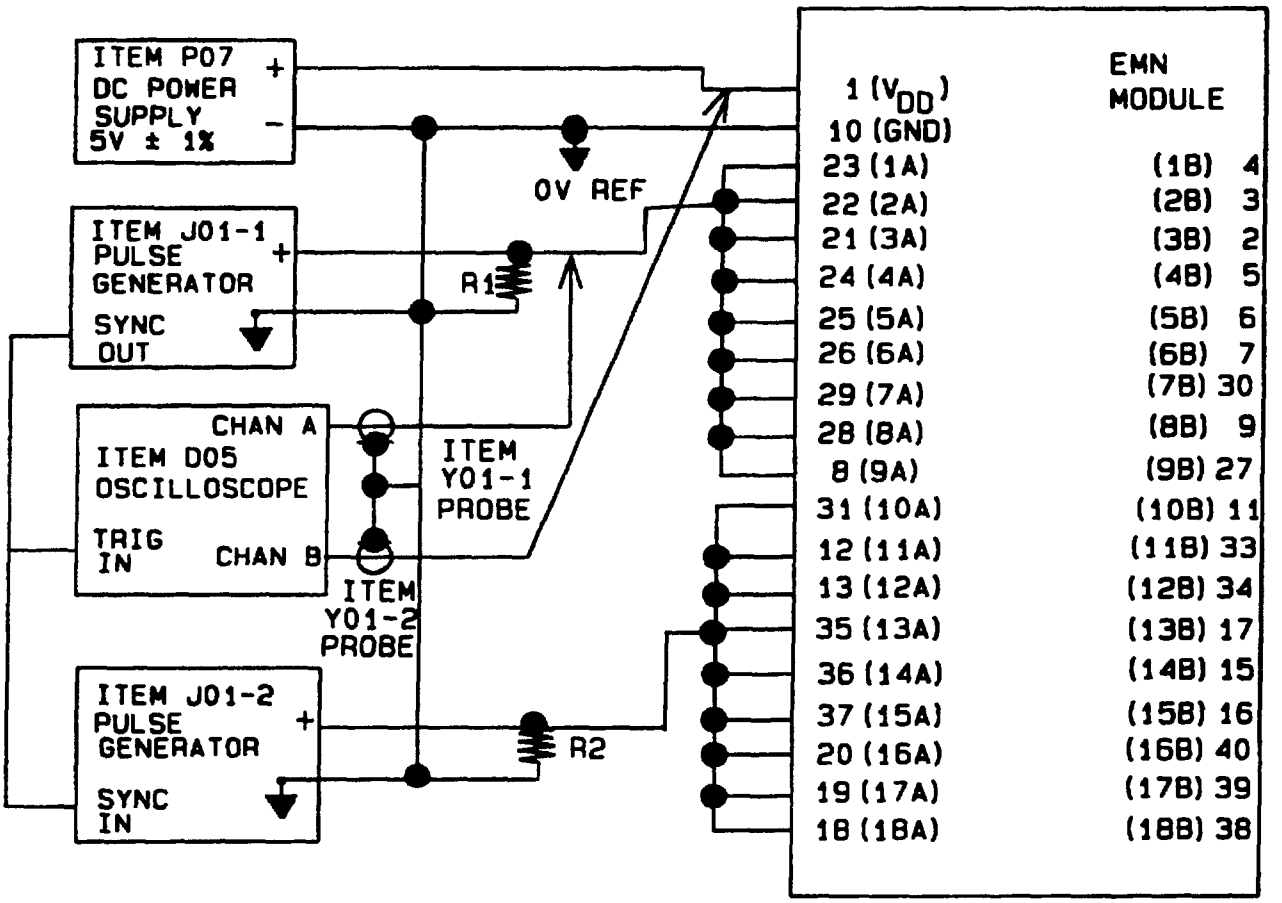
FIGURE 17. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply transient amplitude test.

Test cond	Parameter	Initial limits 25 ± 5°C ambient		Unit	Test fig
		Min	Max		
S	Power supply transient amplitude	0	400	mV	18, 19

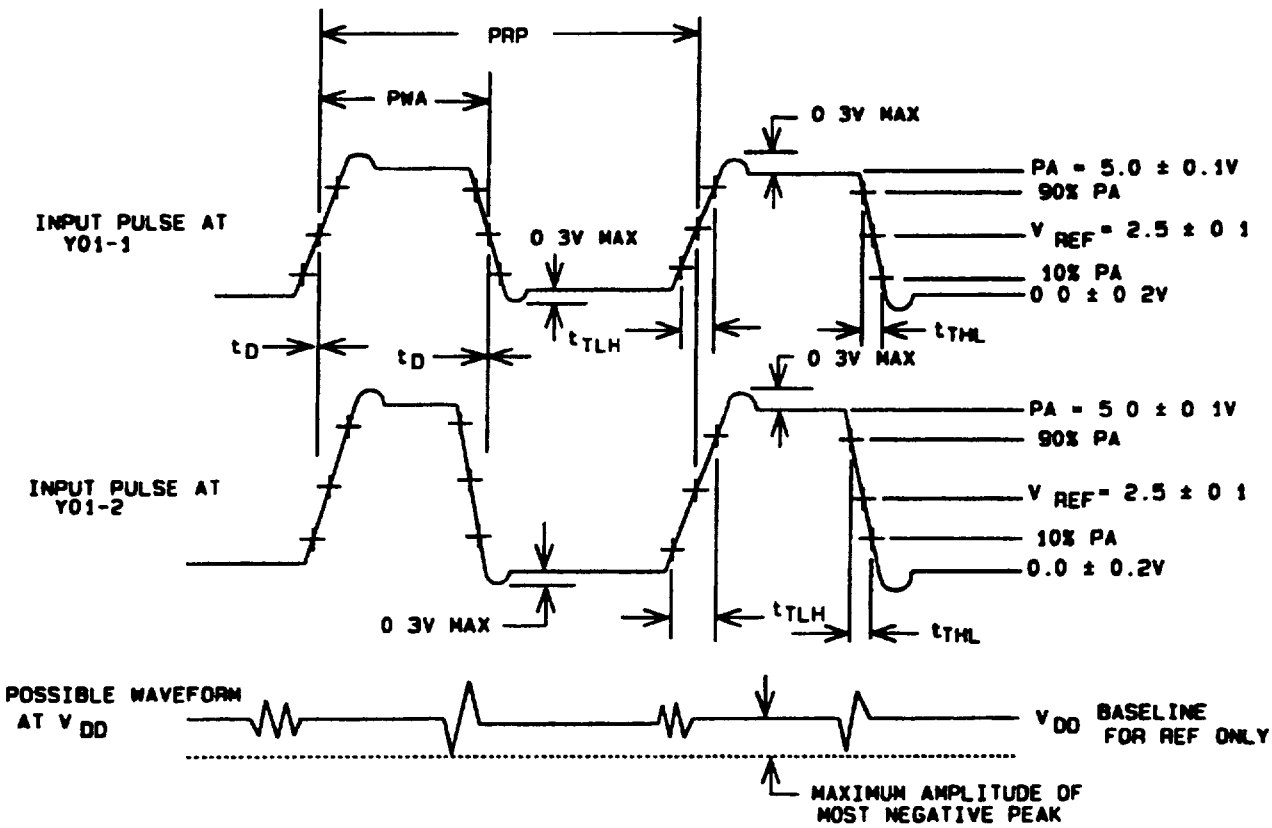
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51.1 Ohms ± 5 per cent, 1/2 W.

* FIGURE 18. Power supply transient amplitude test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:11Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters: (verify the input pulse parameters at each input pin before making the measurement).
PRP = 50 ± 5 µS.
PWA = 25 ± 2.5 µS.
tTLH = tTHL = 20 ± 4 nS.
tD = 0.0 ± 5 nS.

* FIGURE 19. Power supply transient amplitude test circuit waveforms for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XIX. Power supply filter capacitance test.

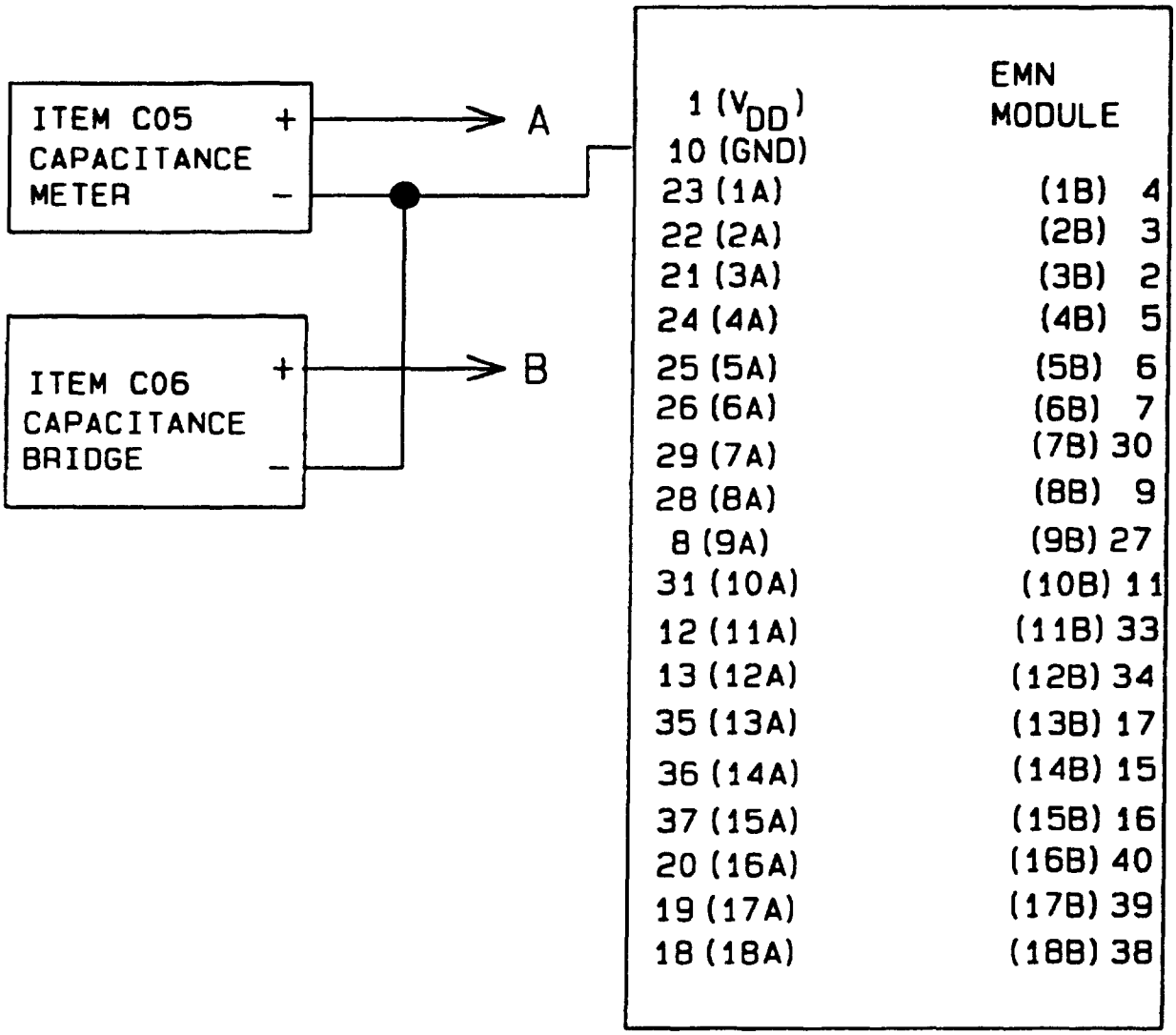
Test cond	Parameter	Connect		Initial limits 25 ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	Open	1	0.007	0.013	μF	20

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:11Z
Check the source to verify that this is the current version before use.

TABLE XX. Input capacitance test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25 ± 5°C ambient Min	Initial limits 25 ± 5°C ambient Max	Unit	Test fig
U1	1A Input capacitance	23	Open	2	25	pF	20
U2	2A	22					
U3	3A	21					
U4	4A	24					
U5	5A	25					
U6	6A	26					
U7	7A	29					
U8	8A	28					
U9	9A	8					
U10	10A	31					
U11	11A	12					
U12	12A	13					
U13	13A	35					
U14	14A	36					
U15	15A	37					
U16	16A	20					
U17	17A	19					
U18	18A Input capacitance	18	Open	2	25	pF	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C06 setup:
   Frequency = 1 kHz.
   Vpk = 0.3 V maximum.

* FIGURE 20. Power supply filter and input capacitance test circuit for tables XIX and XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7
3. The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-M-28787/262A	1994 March 23

3. DOCUMENT TITLE
Modules, Standard Electronic, Inverter, Buffer, Digital, Key Code EMN

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME Technical Point of Contact (TPOC): Mr. Joseph P. Toy (Code 603A)	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-5993	88-482-5993
c ADDRESS (Include Zip Code) Commander ATTN: Joseph P. Toy, Code 603A, Bldg. 2917 NAVSURFWARCENDIV, 300 Highway 361 Crane, IN 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:11Z
Check the source to verify that this is the current version before use.